

GrainFlow

Warehouse Management System

User Manual & Admin Guide

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Chapter 1: Getting Started

This chapter covers how to create your GrainFlow account, log in, set up your warehouse, and find your way around the dashboard.

1.1 Creating Your Account

To start using GrainFlow, you need to create an account.

1. Open GrainFlow in your browser.
2. On the login page, click Sign up.
3. Fill in the following details:
 - Full Name -- your name as it should appear in the system.
 - Email -- your email address (e.g., name@company.com).
 - Password -- choose a password with at least 6 characters. The strength indicator will show Weak, Fair, or Strong as you type.
 - Mobile Number -- your 10-digit mobile number.
4. Click Sign Up.
5. You will be redirected to the dashboard to set up your warehouse.

[See application for visual reference]

Tip: Use a strong password (8+ characters) for better security. The strength bar turns green when your password is strong enough.

1.2 Logging In

Once your account exists, log in as follows:

1. Open the GrainFlow login page.
2. Enter your Email or Mobile Number in the first field.
 - You can type your 10-digit mobile number directly -- no need to add +91.
 - Or use your full email address.
3. Enter your Password.
4. Click Login.

[See application for visual reference]

If you have forgotten your password, click the Forgot password? link below the password field and follow the instructions sent to your email.

Tip: Logging in with your mobile number is the fastest option. Just type your 10 digits and your password.

1.3 Setting Up Your First Warehouse

After signing up for the first time, GrainFlow will ask you to create your warehouse. This is a one-time setup.

1. Enter Warehouse Name (e.g., "Main Godown" or "Sai Storage").
2. Enter Location (e.g., "Guntur, Andhra Pradesh").
3. Enter your Phone number and Email (optional).
4. Enter Total Capacity (Bags) -- this is the maximum number of bags your warehouse can hold.
5. Click Get Started.

[See application for visual reference]

Your warehouse is now ready. You will see the main dashboard.

Important: The capacity you set here is used to show how full your warehouse is on the dashboard. You can update it later in Settings.

1.4 Staff and Team Members

If you are a staff member joining an existing warehouse (not the owner), you will see a Join Warehouse screen instead of the setup screen. Enter the warehouse code provided by your owner or admin to join.

1.5 The Dashboard

The dashboard is the first screen you see after logging in. It gives you a quick overview of your warehouse operations.

[See application for visual reference]

The dashboard is organized into these sections:

Hero Section

At the top, you will see:

- Warehouse Name -- the name of your current warehouse.
- Total Stock -- how many bags are currently stored.
- Capacity -- a visual bar showing how full your warehouse is.
- Active Records -- the number of storage records that are currently open (stock not yet withdrawn).

Quick Shortcuts

Below the hero section, shortcut buttons let you jump directly to common tasks like creating a new inflow, processing an outflow, or adding a customer.

Overview Stats

Key numbers displayed as cards:

- Total stock in the warehouse.
- Number of active storage records.
- Financial summaries (total billed, total collected, pending amounts).

Market Prices

A widget showing current market prices for common agricultural commodities, updated regularly.

Recent Activity

A timeline of the most recent inflows and outflows, showing date, customer name, commodity, and bags.

1.6 Navigation

GrainFlow has a sidebar menu (on desktop) or a bottom navigation bar (on mobile) with the following sections:

Menu Item | What It Does

Dashboard | Overview of your warehouse

Customers | Manage customer profiles and balances

Inflow | Record new stock arrivals

Outflow | Process stock withdrawals and generate bills

Payments | View pending payments and record collections

Expenses | Track warehouse operational costs

Settings | Manage lots, crops, team members, and preferences

To navigate, click the menu item on the sidebar or tap it on mobile.

Tip: The breadcrumb trail at the top of every page shows where you are. For example: Dashboard > Customers > Add Customer. Click any breadcrumb to go back to that level.

1.7 Dark Mode

GrainFlow supports dark mode for comfortable use in low-light conditions.

To switch between light and dark mode:

1. Go to Settings or look for the theme toggle in the top navigation bar.
2. Choose Light, Dark, or System (matches your device setting).

Tip: If you work in a godown with bright sunlight during the day and dim light in the evening, use System mode so GrainFlow switches automatically.

1.8 Subscription Alerts

If your subscription is nearing expiry or has expired, a banner will appear at the top of the dashboard alerting you. Contact your administrator or follow the renewal link in the alert to continue using GrainFlow without interruption.

Next: Chapter 2 -- Customers

Chapter 2: Customers

This chapter covers how to add, view, edit, search, and manage your customer records.

2.1 Viewing the Customer List

To see all your customers:

1. Click Customers in the sidebar menu.

You will see a table (on desktop) or card list (on mobile) showing:

Column	Description
Name	Customer's full name
Phone	Contact number
Village	Customer's village
Active Records	Number of open storage records for this customer
Total Billed	Total amount billed across all records
Paid	Total amount the customer has paid
Balance Due	Outstanding amount (Billed minus Paid)

[See application for visual reference]

2.2 Adding a New Customer

1. Go to Dashboard > Customers.
2. Click the Add Customer button in the top right.
3. Fill in the customer details:
 - Name (required) -- the customer's full name.
 - Father's Name -- helps identify customers with similar names (common in rural areas).
 - Village -- the customer's village. Useful for filtering later.
 - Address (required) -- full address for records and receipts.
 - Phone (required) -- 10-15 digit mobile number. You may include +91 or leave it off.
 - Email (optional) -- if the customer has an email, entering it here lets them access the self-service Customer Portal to view their own stock and payments.
4. Click Save Customer.

[See application for visual reference]

Important: The phone number must be unique. You cannot add two customers with the same phone number.

Tip: Even if a customer does not use email, always enter their phone number accurately. It is used for sending SMS notifications about inflows and outflows.

2.3 Viewing a Customer Profile (360-Degree View)

Click any customer's name in the list to open their full profile. The customer profile page gives you a complete picture:

- Customer Details -- name, father's name, village, phone, address.
- Active Storage Records -- all open records showing commodity, bags stored, lot location, and storage start date.
- Payment History -- a list of all payments made against each record.
- Balance Summary -- total billed, total paid, and outstanding balance at a glance.
- Statement -- generate a PDF statement for the customer.

[See application for visual reference]

From the customer profile, you can:

- Record a payment (click Record Payment).
- Process a bulk outflow (click Bulk Outflow).
- View or download a statement.
- Edit or delete the customer.

2.4 Editing a Customer

1. Open the customer's profile page.
2. Click the Edit button.
3. Update the fields you need to change (name, phone, village, address, etc.).
4. Click Save to apply changes.

2.5 Deleting a Customer

1. Open the customer's profile page.
2. Click the Delete button.
3. Confirm the deletion.

Important: Deleting a customer does not permanently erase them. They are soft-deleted and can be restored later using the Restore Customer option on the Customers page. However, a deleted customer will not appear in dropdowns for new inflows or payments.

To restore a deleted customer:

1. Go to Dashboard > Customers.
2. Click Restore Customer (next to the Add Customer button).
3. Select the customer from the list and confirm.

2.6 Searching and Filtering Customers

Search

Use the search bar at the top of the Customers page to find customers by name. The search runs on the server, so it works even with a large number of customers.

Filters

Click the Filter button to narrow down the list:

- Village -- select one or more villages.
- Balance Range -- set a minimum and/or maximum balance to find customers who owe within a specific range.

Sort

Click the Sort dropdown to order customers by:

- Highest or Lowest Balance
- Name (A-Z or Z-A)
- Most or Least Billed
- Most or Least Active Records

Share Filters

After applying filters, use the Share button to copy a URL with your current filter settings. Share this link with a colleague so they see the same filtered view.

Export

Click Export to download the filtered customer list as an Excel file. The export includes all columns visible in

the table.

[See application for visual reference]

Tip: To quickly find customers with overdue payments, sort by "Highest Balance" and look at the top of the list.

2.7 Customer Statements

A customer statement is a summary document showing all of a customer's storage records, charges, and payments.

To generate a statement:

1. Open the customer's profile page.
2. Click Statement or the statement icon.
3. Optionally select a date range to limit the statement period.
4. Click Download PDF to save the statement to your device.

You can print this PDF and hand it to the customer as a record of their account.

[See application for visual reference]

Next: Chapter 3 -- Storage Inflow

Chapter 3: Storage Inflow

This chapter covers how to record new stock arriving at your warehouse, including assigning it to a storage lot and calculating hamali charges.

3.1 What Is an Inflow?

An inflow is the process of recording new bags of grain or other commodity entering your warehouse for storage. Each inflow creates a storage record -- a unique entry that tracks:

- Which customer the stock belongs to.
- What commodity it is (e.g., paddy, groundnut, maize).
- How many bags were brought in.
- Which lot (storage zone) in the warehouse it was placed in.
- The date storage started.
- Hamali (labor) charges for loading/unloading.

Important: Rent is not charged at inflow time. Rent is calculated only when the customer withdraws their stock (outflow). The inflow simply starts the storage clock.

3.2 Creating a New Storage Record

1. Go to Dashboard > Inflow.
2. The inflow form will appear on the page. Fill in the details:

Step-by-Step

Select Customer

3. Choose the customer from the Customer dropdown. If the customer does not exist yet, click Add Customer in the top right to create one first.
4. Once selected, the customer's Father's Name and Village will auto-fill (read-only) for easy verification.

Choose Inflow Type

5. Select the inflow type:
 - Direct (Purchase) -- stock arriving fresh from the field or market. This is the most common type.
 - Plot (Transfer In) -- stock being transferred from an outdoor plot into the warehouse.

Select Product and Lot

6. Choose the Product / Crop from the dropdown (e.g., Paddy, Groundnut).
7. Choose the Lot No. where the bags will be stored. The dropdown shows available capacity for each lot (e.g., "Lot A (Available: 450)"). Lots that are full are disabled.

Enter Quantities

8. For a Direct inflow, enter the No. of Bags being stored.
9. For a Plot transfer, enter:
 - Plot Bags -- the number of bags coming from the plot.
 - Load Bags (Final) -- the final count after re-weighing (optional).

Vehicle and Date

10. Enter the Lorry / Tractor No. (e.g., "AP 21 1234") -- optional but useful for tracking.
11. Set the Date. It defaults to today but can be changed to a past date if you are recording a delayed entry.

Hamali Charges

12. Enter the Hamali Rate (per bag) -- the rate charged per bag for loading/unloading labor.
13. Enter Hamali Paid Now -- if the customer is paying hamali charges upfront, enter the amount here. Leave blank or 0 if payment is deferred.

Khata Amount

14. If applicable, enter the Khata Amount (Weighbridge) -- the weighbridge charge.

[See application for visual reference]

Billing Summary

At the bottom of the form, a summary shows:

- Total Hamali Payable -- bags multiplied by hamali rate. If linked to an unloading record, includes unloading charges.
- Hamali Pending -- hamali payable minus hamali paid now.
- Estimated Rent (6 Months) -- a reference showing what the per-bag rent would be if withdrawn within 6 months. This is informational only; actual rent is calculated at outflow.

15. Click Create Storage Record.

The record is created and assigned an auto-generated serial number.

3.3 Using Unloading Records

If your warehouse uses the gate entry / unloading workflow, you can link an inflow to a previous unloading record:

1. At the top of the inflow form, you will see a section labeled Select from Unloading Record (Optional) if any unloaded trucks are waiting.
2. Select a truck arrival from the dropdown. It shows the customer name, commodity, and bags remaining.
3. The form will auto-fill the customer, crop, and bag count from the unloading record.
4. Complete the remaining fields and submit.

Tip: Using unloading records speeds up data entry and reduces errors because the customer and commodity are already known.

3.4 Understanding the Receipt

After creating a storage record, a receipt is generated that you can print or download. The receipt includes:

- Record serial number.
- Customer name, father's name, and village.
- Commodity and lot number.
- Number of bags stored.
- Storage start date.
- Hamali charges (total and pending).
- Lorry/tractor number.

Hand the printed receipt to the customer as proof of storage.

[See application for visual reference]

3.5 Viewing Recent Inflows

The Inflow page shows a list of the most recent inflows below the form. You can see:

- Record number.
- Customer name.
- Commodity.

- Bags.
- Date.
- Status (Active or Withdrawn).

Use this list to verify recent entries or find a record to review.

3.6 Tips and Common Issues

Tip: Always double-check the lot selection. Placing bags in the wrong lot makes physical retrieval difficult later.

Tip: If a customer brings stock in multiple lorry loads on the same day, create one inflow record per lorry for accurate tracking.

Important: You cannot change the customer on a storage record after it is created. If you select the wrong customer, you will need to delete the record and create a new one.

Important: The storage start date determines when rent starts accumulating. If you backdate an inflow, the rent calculation at outflow will cover the full period from that date.

Next: Chapter 4 -- Storage Outflow

Chapter 4: Storage Outflow

This chapter covers how to process stock withdrawals, understand rent calculation, apply discounts, collect payment, and handle bulk outflows.

4.1 What Is an Outflow?

An outflow is the process of withdrawing bags from storage. When a customer collects their stock, you process an outflow to:

- Record how many bags are leaving.
- Calculate the rent owed based on how long the bags were stored.
- Collect payment (fully or partially).
- Generate a final bill / receipt.

4.2 Processing a Single Record Outflow

1. Go to Dashboard > Outflow.
2. In the Search Record field, search for the storage record by record number, customer name, or commodity.
3. Select the record from the search results. The system loads the record details and shows the inflow date.

[See application for visual reference]

4. Enter the Bags to Withdraw. The maximum is the number of bags currently stored in that record.
5. Set the Withdrawal Date. It defaults to today.

Billing Summary

Once you enter the bags and date, the system automatically calculates:

- Storage Duration -- how many months the bags were stored (rounded up -- even 1 day into a new month counts as a full month).
- Standard Rent -- the total rent based on tiered rates (see section 4.3 below).
- Discount -- enter a discount amount if applicable. The discount is subtracted from the standard rent.
- Net Rent Due -- standard rent minus any discount.
- Pending Hamali Charges -- any unpaid hamali from the original inflow.

- Prior Credit / Advance -- if the customer has overpaid hamali or has credit, it is deducted here.
- Total Payable -- the final amount the customer owes (net rent + pending hamali - advance).

6. Enter the Total Paid Now -- the amount the customer is paying today. Leave blank if the customer will pay later.

7. Click Process Outflow.

[See application for visual reference]

Important: If you withdraw all remaining bags from a record, the record is automatically closed and marked as "Withdrawn." If you withdraw only some bags, the record stays open with the remaining bag count updated.

4.3 How Rent Is Calculated (Tiered Rates)

GrainFlow uses a tiered pricing model based on storage duration. The default rates are:

Duration | Rate Per Bag

1 to 6 months | Rs. 36 per bag

7 to 12 months | Rs. 55 per bag

These are flat rates for the entire period, not monthly charges. For example:

- 100 bags stored for 3 months = $100 \times \text{Rs. } 36 = \text{Rs. } 3,600$
- 100 bags stored for 9 months = $100 \times \text{Rs. } 55 = \text{Rs. } 5,500$

Important: The rate per bag applies to the full tier. Storing for 1 month costs the same as storing for 6 months (both Rs. 36 per bag). The jump to Rs. 55 happens only if storage exceeds 6 months.

Multi-Year Storage

For storage beyond 12 months, rent is calculated in yearly blocks:

- Year 1 (months 1-12): Rs. 55 per bag (the 1-year rate).
- Each additional year: The rate resets and follows the same tier logic.

Example: 100 bags stored for 14 months (1 year + 2 months):

- Year 1 (12 months): $100 \times \text{Rs. } 55 = \text{Rs. } 5,500$
- Remaining 2 months (falls in the 0-6 month tier): $100 \times \text{Rs. } 36 = \text{Rs. } 3,600$
- Total: Rs. 9,100

Month Rounding Rule

Even 1 day into a new month counts as a full month. For example:

- Storage from January 1 to July 2 = 7 months (not 6), so the 1-year rate of Rs. 55 applies.

Dynamic Pricing

Your warehouse may have custom rates configured per crop. If a crop has custom pricing set up in Settings, GrainFlow uses those rates instead of the defaults shown above. The billing summary always shows the actual rate being applied.

Tip: Review the billing summary carefully before confirming an outflow. Once processed, the rent is recorded on the storage record.

4.4 Applying a Discount

To apply a discount:

1. In the outflow form, after the system shows the Standard Rent, enter the discount amount in the Discount field.
2. The Net Rent Due updates automatically.

Discounts are useful for:

- Long-standing customers.
- Bulk withdrawals.
- Goodwill gestures for damaged goods.

The discount is recorded on the outflow transaction for your records.

4.5 Collecting Payment at Outflow

The Total Paid Now field allows you to record payment at the time of outflow:

- If the customer pays in full, enter the Total Payable amount.
- If the customer pays partially, enter the partial amount. The remainder becomes an outstanding balance on the customer's account.
- If the customer will pay later, leave the field blank or enter 0.

Tip: You can always record additional payments later through the Payments section (see Chapter 5).

4.6 Bulk Outflow (FIFO)

When a customer wants to withdraw a large quantity of bags that spans multiple storage records, use Bulk Outflow instead of processing each record individually.

What Is FIFO?

FIFO stands for First In, First Out. It means the system withdraws from the oldest storage records first. This is the standard practice in warehouse management -- stock that arrived first should leave first.

How to Process a Bulk Outflow

1. Go to the customer's profile page (Dashboard > Customers > click the customer name).
2. Click the Bulk Outflow button.
3. Select the Commodity from the dropdown. Only commodities the customer currently has in storage will appear.
4. Set the Withdrawal Date.
5. Enter the Bags to Withdraw. The system shows the total available bags for that commodity. Click Max to withdraw everything.

[See application for visual reference]

Preview Plan

As soon as you enter a valid bag count, the system shows a preview table:

- Each row represents a storage record that will be affected.
- Records are listed in FIFO order (oldest first).
- For each record, you see:
 - Record Number and inflow date.
 - Current Stock -- bags currently in that record.
 - Withdraw -- how many bags will be taken from this record.
 - Rent -- the calculated rent for the withdrawn bags.
 - CLOSE label if the entire record is being emptied.

You can uncheck any record to exclude it from the bulk withdrawal. This is useful if certain bags should not be touched.

Summary

Above the table, a summary shows:

- Total Bags being withdrawn.

- Standard Rent across all affected records.
- Pending Hamali from those records.
- Prior Credit / Advance if any.
- Total Payable Today -- the grand total.

6. Optionally enter a Discount amount (distributed across records).
7. Optionally enter a Rent Payment amount (distributed proportionally across records).
8. Click Confirm & Process.

Important: Bulk outflow is a powerful operation. Double-check the preview before confirming. Make sure the right records are selected and the bag counts are correct.

Tip: Bulk outflow saves significant time when a customer has many records of the same commodity (e.g., 15 records of paddy accumulated over a season).

4.7 Outflow Receipt

After processing an outflow (single or bulk), a receipt is generated showing:

- Customer name and details.
- Record number(s).
- Commodity and bags withdrawn.
- Storage duration and rate applied.
- Rent charged, discount (if any), hamali charges, and total payable.
- Amount paid and remaining balance.

You can print or download this receipt to hand to the customer.

[See application for visual reference]

4.8 Reversing an Outflow

If an outflow was processed by mistake, you can reverse it:

1. Find the outflow in the Outflow page list or on the customer's profile.
2. Click the Delete (trash icon) button on the outflow entry.
3. Confirm the reversal.

Reversing an outflow:

- Restores the bags to the original storage record.
- Removes the rent charge from the record.
- Reopens the record if it was closed by the outflow.

Important: Only reverse an outflow if the stock was not actually taken by the customer. Reversals affect financial records, so use them carefully.

4.9 Editing an Outflow

If you need to correct the bag count or other details on an existing outflow:

1. Find the outflow transaction.
2. Click the Edit button.
3. Update the fields as needed.
4. Save the changes.

The system recalculates rent and updates the storage record accordingly.

Important: You cannot increase the bags withdrawn beyond what was available in the record at the time. If you need to withdraw more, process a new outflow instead.

Next: Chapter 5 -- Payments

Chapter 5: Payments

This chapter covers how to record payments from customers, view pending balances, allocate payments to specific storage records, and use bulk payment features.

5.1 Understanding Payments in GrainFlow

In GrainFlow, payments are always linked to a specific storage record. When you record a payment, it reduces the outstanding balance on that record.

A customer's total balance is the sum of all charges (rent + hamali) across their records, minus all payments received.

There are two types of payment transactions:

Type	Description
Rent	Payment toward storage rent or other charges on a record
Hamali	Additional hamali (labor) charge added to a record

5.2 Viewing Pending Payments

To see all customers with outstanding balances:

1. Go to Dashboard > Payments (or Payments > Pending in the sidebar).

The Pending Payments page shows a list of customers who owe money, including:

- Customer name.
- Total outstanding balance.

From this page, you can directly record payments for any customer.

[See application for visual reference]

5.3 Recording a Payment Against a Single Record

This is the most common payment workflow:

1. Go to the customer's profile page (Dashboard > Customers > click the customer).
2. Click the Record Payment button.
3. A dialog shows all of the customer's active storage records with their balances:
 - Record number.
 - Inflow date.
 - Commodity.
 - Bags stored.
 - Total billed, total paid, and balance due.
4. Click Pay next to the record you want to apply the payment to.

[See application for visual reference]

5. In the payment form:
 - Type -- select Rent/Other for a regular payment, or Hamali Charge to add an additional hamali charge to the record.
 - Amount -- enter the payment amount. For rent payments, the maximum is the balance due on that record.
 - Date -- defaults to today; change if recording a past payment.
6. Click Record Transaction.

[See application for visual reference]

Tip: The "Pay" button is disabled for records with zero balance. If a record is fully paid, you do not need to record anything.

5.4 Adding Extra Hamali Charges

Sometimes additional hamali charges arise after the initial inflow (e.g., re-stacking, shifting bags within the warehouse). To add them:

1. Follow the payment recording steps above.
2. In the Type selection, choose Hamali Charge.
3. Enter the additional hamali amount.
4. Click Record Transaction.

This adds to the record's total billed amount, increasing the customer's balance.

5.5 Bulk Payment (Paying Across Multiple Records)

When a customer makes a single large payment that should cover multiple records, use Bulk Payment:

1. Open the customer's profile.
2. Click Bulk Payment.
3. Enter the Payment Amount -- the total amount the customer is paying.
4. Set the Payment Date.

Allocation Strategy

Choose how the payment should be distributed:

- Auto (FIFO - Oldest First) -- the system allocates money to the oldest records first, clearing them in order. This is the recommended approach.
- Manual Distribution -- you specify exactly how much goes to each record.

[See application for visual reference]

FIFO Preview

When using Auto (FIFO), a preview table shows:

- Each record with its current balance.
- How much of the payment will be allocated to each record.
- The remaining balance after allocation.
- A green checkmark for records that will be fully paid off.

Manual Distribution

When using Manual, each record shows an input field where you type the specific amount to allocate. The system warns you if the sum of your allocations does not match the total payment amount.

5. Review the allocation preview.
6. Click Process Payment.

Tip: FIFO allocation is faster and clears old records first. Use Manual only when the customer specifically requests payment toward a particular record.

Important: The payment amount cannot exceed the customer's total outstanding dues.

5.6 Payment History

To view a customer's complete payment history:

1. Open the customer's profile page.
2. Scroll to the storage records section. Each record shows its payments.

Each payment entry shows:

- Date.
- Amount.
- Type (rent or hamali).

5.7 Editing a Payment

If a payment was recorded with the wrong amount or date:

1. Find the payment on the customer profile or record detail page.
2. Click the Edit button on the payment entry.
3. Update the amount or date.
4. Save the changes.

5.8 Deleting a Payment

To remove a payment that was entered by mistake:

1. Find the payment entry.
2. Click the Delete button.
3. Confirm the deletion.

Important: Deleting a payment increases the customer's outstanding balance. Only delete if the payment was genuinely recorded in error.

5.9 Razorpay Payment Links

GrainFlow integrates with Razorpay to allow customers to pay online. When this feature is enabled:

- You can generate a payment link and share it with the customer via SMS or WhatsApp.
- The customer clicks the link and pays using UPI, net banking, cards, or other methods.
- Once the payment is confirmed, GrainFlow automatically records it against the customer's account.

This feature requires Razorpay to be configured in your warehouse settings. Contact your administrator for setup.

Tip: Payment links are especially useful for customers who are not physically present at the warehouse. Send them a link and the payment is recorded automatically when they pay.

Next: Chapter 6 -- Expenses

Chapter 6: Expenses

This chapter covers how to track your warehouse's operational expenses, organize them by category, and review your financial position.

6.1 Understanding the Expenses Page

The Expenses page helps you track money going out of your warehouse business -- salaries, fuel, repairs, and other running costs. It is separate from customer-related payments (rent and hamali), which are tracked in the Payments section.

When you open the Expenses page (Dashboard > Expenses), you see three summary cards at the top:

Card | What It Shows

Total Income | Total amount received from all customer payments

Total Expenses | Sum of all recorded operational expenses

Total Balance | Net balance (Income minus Expenses)

Below the summary cards is the expense history list.

[See application for visual reference]

6.2 Adding a New Expense

1. Go to Dashboard > Expenses.
2. Click the Add Expense button in the top right.
3. Fill in the expense details:
 - Date -- the date the expense occurred. Defaults to today.
 - Category -- select one of the predefined categories:
 - Worker Salary -- wages paid to warehouse workers.
 - Petrol -- fuel costs for generators, vehicles, etc.
 - Maintenance -- repairs, painting, equipment servicing.
 - Utilities -- electricity, water, internet bills.
 - Hamali -- labor charges paid to outside workers (not linked to a specific customer record).
 - Other -- any expense that does not fit the above categories.
 - Description -- a short note about the expense (e.g., "Generator diesel for March" or "Roof leak repair").

- Amount -- the amount in Rs.

4. Click Save Expense.

[See application for visual reference]

Tip: Be specific in the description field. Writing "Petrol - 20 litres for generator" is more useful than just "Petrol" when you review expenses later.

6.3 Viewing Expense History

The expense list below the summary cards shows all recorded expenses. For each expense, you can see:

- Date.
- Category.
- Description.
- Amount.

Filtering and Searching

Use the search and filter controls to narrow down expenses:

- Search -- type a keyword to find expenses by description.
- Filter by Category -- select a category to see only those expenses.
- Filter by Date -- set a date range to view expenses for a specific period.

Tip: At the end of each month, filter expenses by the current month to get a quick summary of your operating costs.

6.4 Editing an Expense

If you entered the wrong amount, category, or description:

1. Find the expense in the list.
2. Click the Edit option from the actions menu (three dots icon).
3. Update the fields as needed.
4. Click Save to apply changes.

6.5 Deleting an Expense

To remove an expense that was entered by mistake:

1. Find the expense in the list.
2. Click the Delete option from the actions menu.
3. Confirm the deletion.

Important: Deleting an expense changes your Total Expenses and Total Balance figures. Only delete if the entry was genuinely a mistake.

6.6 Understanding the Financial Summary

The three cards at the top give you a real-time picture of your warehouse finances:

- Total Income (green) -- all money received from customers (rent payments, hamali payments, outflow collections).
- Total Expenses (red) -- all operational costs you have recorded.
- Total Balance -- Income minus Expenses. If this number is positive, your warehouse is profitable. If negative, expenses have exceeded income.

Example:

- Total Income: Rs. 2,50,000
- Total Expenses: Rs. 1,80,000
- Total Balance: Rs. 70,000 (profit)

Tip: Review your financial summary regularly. If the balance is shrinking month over month, look at the expense categories to see where costs are increasing.

6.7 Best Practices for Expense Tracking

1. Record expenses daily. Do not wait until the end of the month. Daily recording is more accurate and prevents forgotten entries.
2. Use the correct category. This makes filtering and reporting meaningful. If a category does not fit, use "Other" and write a clear description.
3. Include all costs. Even small expenses (e.g., Rs. 50 for tea for laborers) add up over time. Tracking them gives you a true picture of your costs.

4. Review monthly. At the end of each month, filter to that month and review total expenses by category. Look for unusual spikes.

Previous: [Chapter 5 -- Payments](#)

Chapter 7: Reports

Generate financial and operational reports to track revenue, monitor stock, and keep your records audit-ready.

Overview

GrainFlow provides three categories of reports, accessible from Reports in the sidebar:

- Financial Analytics -- Revenue, collections, outstanding dues, and expense tracking
- Operational Analytics -- Warehouse capacity, turnover, commodity breakdown, and customer behaviour
- Custom Reports -- On-demand reports with date ranges, filters, and PDF/Excel export

[See application for visual reference]

7.1 Financial Analytics

Navigate to Reports > Financial Analytics.

The financial dashboard is organized into five tabs:

Overview Tab

This is the default view. It shows four headline metrics at the top:

Metric | What It Means

Total Revenue | All rent + hamali charges billed across all records

Total Collected | Amount actually received from customers

Outstanding Dues | Billed amount that has not been paid yet

Avg. Days to Payment | How long customers typically take to pay

Below the metrics you will find:

1. Revenue Trends (Last 12 Months) -- A line chart breaking down monthly Rent vs. Hamali revenue. A dashed line shows the combined total.
2. Top 10 Customers by Revenue -- A ranked list of your highest-revenue customers, showing amount paid and any outstanding balance. Click a name to jump to that customer's profile.
3. Outstanding Dues Aging -- A pie chart showing how old your unpaid dues are (for example, 0-30 days, 31-60 days, 61-90 days, 90+ days). Use this to prioritize collections.

[See application for visual reference]

Unloading (Ops) Tab

A table listing all unloading activities: date, customer name, commodity, lorry number, bags unloaded, and hamali amount charged.

Hamali (Rev) Tab

Hamali revenue grouped by customer. Click a customer row to expand and see individual transactions with dates, bag counts, billed amounts, and balance due.

Receivables Tab

A per-customer breakdown of rent and hamali dues:

- Rent Billed / Rent Paid / Rent Due
- Hamali Billed / Hamali Paid / Hamali Due
- Total Pending

This is the report to check when following up on outstanding payments.

Expenses Tab

A log of unloading-related expenses with date, description, and amount.

7.2 Operational Analytics

Navigate to Reports > Operational Analytics.

Four headline metrics appear at the top:

Metric | What It Means

Capacity Utilization | Percentage of total warehouse space currently in use

Turnover Rate | Ratio of completed outflows to total inflows

Avg. Storage Duration | Average number of days items stay in your warehouse

Repeat Customer Rate | Percentage of customers who have stored more than once

Below the metrics:

1. Lot-wise Capacity Utilization -- A bar chart comparing current stock against total capacity for each lot/zone. Use this to spot lots that are full or underused.

2. Commodity Distribution -- A pie chart of the top 6 commodities by total bags stored, with average storage duration per commodity.

3. Customer Insights -- Visual breakdown of active vs. total customers, repeat rate, and average bags per transaction.

[See application for visual reference]

7.3 Custom Reports

Navigate to Reports > Custom Reports.

The custom report generator lets you build specific reports for compliance, audits, or daily operations.

Available Report Types

Report Type | What It Contains | Date Range Needed?

All Customers List | Every registered customer with active bag counts | No

Customer Dues Statement | Detailed rent + hamali dues for one customer | No (customer selection needed)

Active Inventory | All items currently stored in the warehouse | No

Pending Dues List | Customers with outstanding balances | No

Inflow Register | Items received during a date range | Yes

Outflow Register | Items withdrawn during a date range | Yes

Payment Register | Payments received during a date range | Yes

Lot Inventory | Mapping of lots to customers and items in stock | No

Recent Transactions | Last 1,000 inflow/outflow entries | No

Generating a Custom Report

1. Go to Reports > Custom Reports.
2. Select a Report Type from the dropdown.
3. If the report requires a date range, set the From Date and To Date.
4. If using the Customer Dues Statement, select the specific customer. You can also choose whether to show All dues or Hamali Only, and whether to include settled/closed records.
5. Choose the Export Format: PDF Document or Excel Spreadsheet.
6. Click Generate Report.
7. The file downloads to your device automatically.

[See application for visual reference]

Important: Custom reports require the Starter plan or above. Free plan users will see a message asking

them to upgrade.

7.4 Exporting Data

Exporting from Financial Analytics

1. Open Reports > Financial Analytics.
2. Switch to the tab you want to export (Overview, Unloading, Hamali, Receivables, or Expenses).
3. Click the Export Data button in the top-right corner.
4. An Excel file containing that tab's data will download.

Export Format Details

- Excel exports include formatted columns, headers, and totals where applicable.
- PDF exports from Custom Reports include the warehouse name, report title, and generation timestamp.

Tip: Use the Inflow Register and Outflow Register exports as supporting documents during APMC inspections or audits. The date-range filter makes it easy to match a specific inspection period.

Important: The Export button only appears if your subscription plan includes export access (Starter plan and above). On the Free plan, you will see a note that says "Export unavailable on current plan."

Tips for Using Reports Effectively

- Check Receivables weekly. Sort mentally by total pending to focus collection efforts on the largest outstanding amounts.
- Review Aging monthly. If the 90+ days segment is growing, tighten your payment follow-up process.
- Use Lot Utilization before accepting new stock. It shows exactly which zones have space.
- Export before month-end. Generate the Payment Register for the full month to reconcile with your bank statements.

Chapter 8: Analytics Dashboard

Understand your warehouse performance at a glance with real-time charts and business metrics.

Overview

GrainFlow offers two levels of analytics:

1. Dashboard home page -- Quick summary cards visible every time you log in.
2. Business Analytics page -- Detailed charts for financial performance, stock movement, and year-over-year trends.

8.1 Dashboard Metrics

When you log in, the top of your dashboard shows four summary cards:

Card | What It Shows

Total Stock | Number of bags currently stored, with the occupancy percentage of your total warehouse capacity

Active Records | Count of storage records that are currently open (goods still in the warehouse)

Available Space | Remaining capacity in bags -- how much more you can accept

Pending Revenue | Total outstanding dues (Rs.) across all customers

[See application for visual reference]

Reading the Numbers

- Occupancy Rate is calculated as: $(\text{Total Stock} / \text{Total Capacity}) \times 100$. If this is above 90%, the card turns red to alert you that space is running low.
- Active Records represents current customers with goods in storage. This is not the same as total customers -- it only counts open records.
- Pending Revenue is the sum of all unpaid rent and hamali charges. A rising number here means you need to follow up on collections.

Tip: The dashboard metrics refresh automatically. You do not need to reload the page to see updated numbers after recording an inflow or outflow.

8.2 Business Analytics Page

Navigate to Analytics in the sidebar for the full analytics suite.

Important: The Business Analytics page is available on the Professional plan and above. Free and Starter plan users will not see this option.

Revenue vs. Expenses Chart

A line chart showing monthly revenue (green) and expenses (red) for the current year. This helps you see whether your warehouse is profitable month by month.

[See application for visual reference]

Year-over-Year Growth

A bar chart comparing annual revenue across the last five years. Use this to spot long-term growth trends or seasonal dips.

Net Profit Chart

A bar chart of monthly net earnings (revenue minus expenses). Months where you spent more than you earned will show as shorter or negative bars.

Inflow vs. Outflow Chart

A bar chart comparing monthly bag inflow (blue) against bag outflow (amber) for the current year. This reveals:

- Seasonal patterns -- Which months see the most stock coming in?
- Storage velocity -- Is stock moving out as fast as it comes in?
- Accumulation risk -- If inflow consistently exceeds outflow, your warehouse may fill up.

[See application for visual reference]

8.3 Understanding the Numbers

What "Healthy" Looks Like

Metric | Healthy Range | Action If Outside Range

Occupancy Rate | 60-85% | Below 60%: consider marketing. Above 85%: plan for overflow or new lots.

Turnover Rate | Above 50% | Low turnover means stock is sitting too long. Follow up with customers.

Avg. Storage Duration | Depends on crop | If much higher than typical for the commodity, investigate.

Collection Rate | Above 80% | Below 80%: tighten payment terms, send reminders, check Receivables report.

Common Questions

Q: Why does my Pending Revenue keep going up even though customers are paying?

A: New records generate new rent charges daily. If inflow exceeds payment speed, pending revenue will rise. Check the Aging report (Chapter 7) to ensure old dues are being cleared.

Q: The occupancy rate seems wrong. How is capacity calculated?

A: Total capacity is the sum of all lot capacities configured in Settings > Lots. If you added storage space but did not update lot capacity, the rate will appear inflated. See Chapter 10 for how to update lot capacity.

Q: What time period do the dashboard metrics cover?

A: Dashboard cards show current, real-time state -- not a date range. Active Records means records open right now. Pending Revenue is the total outstanding balance right now.

Tips

- Check the dashboard first thing each morning. The four cards give you a 10-second health check of your warehouse.
- Use Inflow vs. Outflow to plan staffing. If a particular month always sees high inflow, schedule extra labour for unloading.
- Compare YoY Growth before renewing leases or expanding. The trend line shows whether your business justifies the investment.

Chapter 9: Billing & Subscriptions

Manage your GrainFlow subscription plan, view payment history, and upgrade when your warehouse needs grow.

Overview

GrainFlow uses a subscription model with four tiers. All paid plans include a 14-day free trial. You can manage your subscription from Billing in the sidebar.

9.1 Subscription Plans

Feature	Free	Starter (Rs.499/mo)	Professional (Rs.1,499/mo)	Enterprise
Storage Records	250 active	10,000 active	Unlimited	Unlimited
Team Members	2	5	15	Unlimited
Basic Inventory (Inflow/Outflow)	Yes	Yes	Yes	Yes
Customer Management	Yes	Yes	Yes	Yes
Advanced Reports (PDF/Excel export)	--	Yes	Yes	Yes
SMS Notifications	--	Yes	Yes	Yes
Analytics Dashboard	--	--	Yes	Yes
Multiple Warehouses	--	--	Yes	Yes
API Access & Custom Integrations	--	--	--	Yes
Dedicated Account Manager	--	--	--	Yes
Priority Support (24/7)	--	--	--	Yes

Yearly Billing

All plans offer a yearly option at approximately 20% off the monthly price:

- Starter Yearly: Rs.4,790/year (saves roughly Rs.1,198)
- Professional Yearly: Rs.14,390/year (saves roughly Rs.3,598)

Enterprise pricing is custom. Contact the sales team for a quote.

What Counts as a "Storage Record"?

A storage record is one active grain deposit from a customer. For example, if 100 customers each have bags

currently stored, that counts as 100 records. Completed or withdrawn records do not count toward your plan limit.

[See application for visual reference]

9.2 Viewing Your Current Plan

1. Go to Billing in the sidebar.
2. The Current Plan card shows:
 - Your plan name and status (active, trialing, etc.)
 - Next renewal date
 - Usage bar showing how many storage records you have used out of your plan limit

[See application for visual reference]

9.3 Upgrading Your Plan

1. Go to Billing.
2. Click the Upgrade Plan button.
3. A dialog opens showing available upgrade options with prices and feature lists.
4. Click Upgrade to [Plan Name].
5. A secure Razorpay payment link opens in a new tab.
6. Complete payment using UPI, card, net banking, or any supported method.
7. Once payment is confirmed, your plan upgrades immediately.

Tip: Upgrades take effect right away. You get instant access to the new plan's features and limits.

Important: Downgrades apply at the end of your current billing cycle. Your existing features remain available until then.

9.4 Payment Methods

GrainFlow uses Razorpay for secure payment processing. Supported methods:

- UPI (Google Pay, PhonePe, Paytm, etc.)
- Credit and Debit Cards

- Net Banking
- Wallets

You do not need to store card details in GrainFlow. Each payment is processed through a one-time secure link.

9.5 Payment History

1. Go to Billing.
2. The Payment History card on the right shows your recent transactions.
3. Each entry displays:
 - Plan name
 - Payment date
 - Payment method (UPI, Card, etc.)
 - Amount (Rs.)
 - Status (success or failed)

The list shows the five most recent payments. If you have more, a note at the bottom indicates the total count.

[See application for visual reference]

9.6 Invoices

1. Go to Billing.
2. Click View Invoices to see a full list of billing documents.

9.7 Trial and Renewal

14-Day Free Trial

When you sign up for any paid plan, you get 14 days of full access at no cost. No credit card is required to start the trial.

After 14 days:

- If you add a payment method and pay, your subscription continues.
- If you do not pay, your account reverts to the Free tier. Your data is preserved, but features above the Free limit become unavailable.

Renewal

Paid subscriptions renew automatically. Before renewal:

- You will see your next renewal date on the Billing page.
- If payment fails, you will be notified and given a grace period to update payment.

30-Day Refund Policy

If you are not satisfied with a paid plan, contact support within 30 days of payment for a full refund.

Tips

- Start with Free to learn the system. You can always upgrade later without losing any data.
- Choose Yearly billing if you plan to use GrainFlow long-term. The 20% discount adds up over a year.
- Watch the usage bar. If you are approaching your record limit, plan your upgrade before you hit the cap -- you will not be able to add new records once the limit is reached.

Chapter 10: Settings

Configure your warehouse details, manage storage lots, and control team access.

Overview

The Settings page is organized into tabs. Navigate to Settings in the sidebar to access:

- Profile -- Your personal account details
- Warehouse -- Warehouse name, location, contact, GST, and capacity
- Crops & Rates -- Commodity types and rental rate configuration
- Data -- Data management and export tools
- Notifications -- Alert preferences
- SMS -- SMS notification settings
- Billing -- Quick link to subscription management

This chapter covers the Warehouse, Lots, and Team sections in detail.

[See application for visual reference]

10.1 Warehouse Details

Viewing Warehouse Information

1. Go to Settings.
2. Click the Warehouse tab.
3. The form shows your current warehouse details.

Updating Warehouse Details

1. Go to Settings > Warehouse.
2. Edit any of these fields:

Field	Description	Example
-------	-------------	---------

Warehouse Name	The display name used on reports and receipts	Srilakshmi Warehouse
----------------	---	----------------------

Location	Physical address or area	Kurnool, Andhra Pradesh
----------	--------------------------	-------------------------

Phone Number | Contact number for the warehouse | 9703503423

Email | Contact email address | contact@warehouse.com

GST Number | Your GSTIN for tax compliance | 37AABCU9603R1ZM

Capacity (Bags) | Total storage capacity in bags | 50000

3. Click Save Changes.

Important: The warehouse name and GST number appear on printed receipts and PDF reports. Make sure they are accurate.

Switching Between Warehouses

If you manage multiple warehouses (Professional plan and above), a warehouse switcher appears at the top of the Warehouse tab. Select a different warehouse to view and edit its settings.

[See application for visual reference]

10.2 Warehouse Lots (Storage Zones)

Lots are physical zones or compartments within your warehouse. Setting up lots helps you track where specific goods are stored and monitor space utilization per zone.

Viewing Lots

1. Go to Settings > Warehouse tab.
2. Under Lot Configuration, click Manage Lots.
 - Or navigate directly to Settings > Lots from the breadcrumb.
3. You will see a table listing all lots with:
 - Lot Name
 - Capacity (in bags)
 - Current utilization (a progress bar showing how full the lot is)
 - Status

[See application for visual reference]

Creating a New Lot

Adding one lot at a time:

1. Go to Settings > Lots.
2. Click Add Lot.

3. Enter the lot name (for example, "Zone A" or "Godown 1").
4. Enter the capacity in bags.
5. Click Save.

Adding multiple lots at once:

1. Click Bulk Add on the Lots page.
2. Enter the number of lots and a naming pattern.
3. Click Create.

Pasting a lot list:

1. Click the paste option on the Lots page.
2. Paste a list of lot names (one per line).
3. Confirm to create all listed lots.

Editing a Lot

1. Find the lot in the table.
2. Click the Edit button (pencil icon) next to it.
3. Update the name or capacity.
4. Click Save.

Deleting a Lot

1. Find the lot in the table.
2. Click the Delete button next to it.
3. Confirm the deletion.

Important: Deleting a lot does not delete the storage records assigned to it. However, those records will no longer display a lot name. Reassign records to another lot before deleting if possible.

Understanding Utilization

Each lot shows a progress bar:

- Green bar -- The lot is within capacity.
- Red bar -- The lot has exceeded its configured capacity. This means more bags are assigned to it than it should hold. Reduce stock or increase the lot's capacity.

Tip: Set lot capacities to match their real physical limits. This makes the utilization indicators accurate and helps you avoid overfilling a zone.

10.3 Team Management

Team management lets you add staff members who can access your GrainFlow account with controlled permissions.

Accessing Team Settings

1. Go to Settings.
 2. Click the Manage Team button in the top-right corner.
- Or navigate to Settings > Team directly.

The team page shows a two-panel layout:

- Left panel: List of all team members
- Right panel: Details of the selected member, or the add-member form

[See application for visual reference]

Roles Explained

Role | What They Can Do

Staff | Manage storage records (inflow, outflow, payments). Cannot access settings or team management.

Manager | Everything Staff can do, plus manage team members below their level.

Admin | Full access to all features, settings, and team management.

Owner | Account owner. Highest access level. Cannot be removed by others.

Adding a New Team Member

1. Go to Settings > Team.
2. Click Add Member at the bottom of the member list.
3. Fill in the form:
 - Full Name -- The person's display name
 - Email -- Their email address (used to log in)
 - Role -- Staff, Manager, or Admin
 - Assign to Warehouse -- Which warehouse they can access
 - Initial Password -- A temporary password (they can change it after first login)
4. Click Create User.

The new member can now log in with their email and the password you set.

[See application for visual reference]

Using Invite Links

As an alternative to creating accounts directly, you can generate an invite link:

1. Go to Settings > Team > Add Member.
2. Scroll down to the Invite Link section.
3. Click Generate Invite Link.
4. Share the link with the person (via WhatsApp, SMS, etc.).
5. They click the link and complete registration.

Viewing Member Details

1. Click any member in the left panel.
2. The right panel shows their:
 - Name and role
 - Email address
 - Account creation date
 - Current status (active or suspended)
 - Warehouse access

Editing a Team Member

1. Select the member.
2. Click Edit Profile.
3. Update their name, role, or warehouse assignment.
4. Save changes.

Important: You can only edit members who have a lower role than yours. Admins cannot edit other Admins. Only Owners can edit Admins.

Removing Access

1. Select the member.
2. Click the deactivate option.
3. Confirm the action.

The member's account is suspended. They can no longer log in. Their past activity and records remain in the system.

Handling Join Requests

If someone uses an invite link to request access, their request appears as a yellow notification at the top of

the team panel. You can:

- Click Accept to add them to your warehouse.
- Click the X button to reject the request.

Plan Limits for Team Members

The number of team members you can add depends on your plan:

Plan | Maximum Team Members

Free | 1 (owner only)

Starter | 3

Professional | 10

Enterprise | 50

If you have reached your plan's limit, you will need to upgrade before adding more members. See Chapter 9 for upgrade instructions.

Tips

- Set up lots before your first inflow. When you assign incoming stock to a lot, the utilization tracking starts working immediately.
- Use descriptive lot names. Names like "Godown 1 - Left Side" are more useful than "Lot A" when directing labourers.
- Give staff the minimum role needed. Most warehouse workers only need the Staff role. Reserve Admin for people who manage billing and settings.
- Review team access quarterly. Remove accounts for people who no longer work at the warehouse.

Chapter 11: Customer Portal

Give your customers self-service access to view their storage records, payment history, and outstanding balance.

Overview

The Customer Portal is a separate, read-only interface designed for your customers. It lets them check on their stored goods and account balance without calling or visiting your office.

Customers access the portal through a dedicated login page using their registered mobile number. They cannot make changes -- the portal is view-only.

11.1 What Is the Customer Portal?

The portal is a simple, mobile-friendly page that shows a customer:

- Which warehouses hold their goods
- What commodities are stored and in what quantity
- How long each item has been in storage
- How much has been billed and paid
- Any outstanding balance (amount due)
- History of completed (withdrawn) records

Think of it as a live statement that customers can check anytime from their phone.

[See application for visual reference]

11.2 How Customers Access the Portal

Login Process

1. The customer opens the portal URL (for example, yoursite.com/portal/login).
2. They enter their 10-digit mobile number (the one registered in your system).
3. They tap Get Login OTP.

4. An OTP is sent to their phone.
5. They enter the OTP to sign in.

No password is needed. The portal uses secure, one-time codes for authentication.

[See application for visual reference]

Important: The mobile number must match exactly what you entered when creating the customer record in GrainFlow. If a customer cannot log in, verify their phone number in your customer list.

Tip: Share the portal link with customers via SMS or WhatsApp after their first inflow. This saves you from fielding daily "how many bags do I have?" phone calls.

11.3 What Customers Can See

Global Portfolio Header

At the top, a summary card shows:

- Total bags currently stored across all your warehouses
- Number of warehouse locations where they have goods

Active Records

Under the "Active" tab, each warehouse location shows:

- Warehouse name and location
- Total bags in stock at that location
- Amount paid so far (Rs.)
- Amount due (Rs.) -- highlighted in red if there is an outstanding balance
- Per-record details:
 - Commodity name (for example, Groundnut, Paddy)
 - Record number
 - Storage start date
 - Days in storage
 - Number of bags
 - Due amount for that specific record

Customers can tap View Details on any record to see a timeline of all activity:

- Initial inflow (date and bag count)

- Payments received (date and amount)
- Stock withdrawals (date and bags withdrawn)

[See application for visual reference]

History (Completed Records)

The "History" tab shows records where all goods have been withdrawn. Each entry is marked as "Withdrawn" and displays the commodity, dates, and final billing summary.

Download Statement

Customers can tap the Statement button on the portfolio header to download a consolidated PDF of all their records across all warehouses.

Print Receipt

Each record has a Print button that generates a receipt document for that specific storage entry.

11.4 What Customers Cannot Do

The portal is strictly read-only. Customers cannot:

- Add or modify storage records
- Record payments
- Request outflows or withdrawals
- Change their account details
- Access other customers' data
- View warehouse settings or team information

All data changes must be made by your warehouse staff through the main GrainFlow dashboard.

11.5 Admin Access to the Portal

If you (as an admin, manager, or owner) visit the portal page, you will see a Back to Dashboard link that takes you back to the main GrainFlow interface. This is useful when you want to preview what a customer sees.

Regular customers will not see this link.

11.6 Setting Up the Portal for Your Customers

There is no separate setup needed. The portal is always available. To start using it:

1. Make sure your customers have a valid mobile number in their customer record.
2. Share the portal URL with them.
3. They log in with their mobile number and OTP.

That is all. The portal automatically shows data from all warehouses where the customer has records.

Tips

- Print the portal URL on your physical receipts. This helps customers discover the self-service option.
- Tell customers about the Statement download. It gives them a PDF record of everything, which they may need for their own accounting or loan applications.
- If a customer sees "No Records Found" after logging in, check that their phone number in GrainFlow matches the number they used to log in. Even a single digit difference will prevent the match.
- Use the portal as a trust-builder. Customers who can verify their own records are more likely to continue storing with you.

Chapter 20: Admin Dashboard

The Admin Panel is the central control surface for super_admin and owner users. It provides a system-wide view of all warehouses, users, subscriptions, and platform activity. Access it from the sidebar under Admin Panel, or navigate directly to /admin.

Access Requirements

Only users with the super_admin or owner role on their profile can access the Admin Panel. All other roles receive a 403 Forbidden page. This check runs server-side before any data is loaded.

Dashboard Overview

When the Admin Panel loads, the top section displays five summary cards:

Card | Description

Total Warehouses | Count of active tenants on the platform

Registered Users | System-wide user accounts

Total Customers | Farmers and businesses across all warehouses

Active Records | Storage records currently in storage (not withdrawn)

Total Stock | Aggregated bag count across the entire platform

Below the stats cards, the dashboard is organized into six tabs: Warehouses, Users, Audit Logs, Analytics, Subscriptions, and Codes.

Warehouses Tab

The Warehouses tab lists every warehouse on the platform with their name, location, and metadata.

Creating a New Warehouse

1. Click the Create Warehouse button in the page header (top-right).
2. In the dialog, fill in:
 - Name -- the warehouse display name
 - Location -- physical address or city
 - Capacity -- maximum storage capacity in bags
 - GST Number -- optional tax identifier
3. Click Create.

The new warehouse is created immediately. An owner must then be assigned through the Users tab or through the team settings of that warehouse.

Warning: Creating a warehouse does not automatically assign a subscription plan. The warehouse starts on the Free tier with limited capacity. See Chapter 21 for subscription activation.

Editing a Warehouse

Click the edit icon on any warehouse row to modify its name, location, capacity, or GST number. Changes take effect immediately and are reflected across all users of that warehouse.

Users Tab

The Users tab displays a directory of all registered users across the platform. Each row shows the user's name, email, role, and associated warehouse.

Viewing User Details

The table shows each user's:

- Full name and email address
- Global profile role (super_admin, owner, admin, manager, staff)
- Warehouse assignment(s)
- Account creation date

Deactivating a User

User deactivation is handled through warehouse assignment management. To remove a user's access:

1. Navigate to the warehouse's Settings > Team page.
2. Find the user in the team list.
3. Click the toggle to revoke warehouse access.

This performs a soft delete on the warehouse_assignments record (sets deleted_at), preserving the assignment history for audit purposes. The user's authentication account remains intact but they lose access to warehouse data.

Warning: Revoking the last owner's access to a warehouse can leave it without an administrator. Always ensure at least one owner remains assigned.

Audit Logs Tab

The Audit Logs tab displays the Global Audit Log -- a chronological feed of all significant actions across every warehouse on the platform.

Log Entry Fields

Each audit log entry records:

Field | Description

Action | CREATE, UPDATE, DELETE, LOGIN, LOGOUT, EXPORT, BULK_ACTION

Entity | STORAGE_RECORD, CUSTOMER, PAYMENT, INFLOW, OUTFLOW, USER, SETTINGS, SUBSCRIPTION

Entity ID | The specific record that was affected

User | Who performed the action

Warehouse | Which warehouse the action occurred in

IP Address | The IP address of the actor

Timestamp | When the action occurred

Details | A JSON object with action-specific context

Filtering Logs

The audit log supports two filtering mechanisms:

- Search (q): Free-text search across log entries.
- Type filter: Filter by importance level. The default filter is important, which shows only significant operations.

Results are paginated at 50 entries per page. Use the page navigation at the bottom to browse older entries.

Audit Log Retention

Audit logs are append-only and immutable. They cannot be edited or deleted through the application. The `audit_logs` table has no Update type defined in the schema -- entries are write-once.

Analytics Tab

The Analytics tab provides platform-wide metrics and visual charts covering:

- Warehouse growth trends
- User registration patterns
- Storage utilization across the platform
- Revenue and subscription distribution

This data is sourced from the `getPlatformAnalytics()` query and rendered in the Analytics Section component with interactive charts.

Subscriptions Tab

See Chapter 21: Subscription Management for full details on this tab, including plan management, manual activation, and subscription code generation.

Codes Tab

The Codes tab manages activation codes for manual subscription distribution. Super admins can:

- Generate new activation codes tied to specific plans
- View all codes with their status (available, used, revoked)
- See which warehouse redeemed each code

See Chapter 21 for the complete code generation workflow.

Chapter 21: Subscription Management

GrainFlow uses a tiered subscription model to control feature access and usage limits per warehouse. The `super_admin` manages all subscriptions through the Admin Panel's Subscriptions and Codes tabs.

Plan Tiers and Limits

The platform defines four plan tiers. Each tier controls two primary limits and gates access to specific features:

Usage Limits

Resource		Free		Starter		Professional		Enterprise
Storage Records		100		10,000		100,000		1,000,000
Team Members		1		3		10		50

Feature Gates

Feature		Minimum Tier
Advanced Reports Export		Starter
WhatsApp Notifications		Starter
Analytics Dashboard		Professional
Multiple Warehouses		Professional
API Access		Enterprise

Plans can be billed monthly or yearly (e.g., `starter_monthly`, `professional_yearly`). The billing period does not affect limits -- only the pricing and renewal cycle differ.

Subscription Statuses

Each subscription has a status that determines access level:

Status		Meaning
active		Paid and operational. Full access to plan features.
incomplete		Subscription request logged but not yet paid or activated.
trialing / trailing_trial		Trial period active. Full feature access.
past_due		Payment overdue. Features still accessible temporarily.
grace_period		Subscription expired. 7-day window to renew before downgrade.
expired		Grace period ended. Queued for downgrade to Free.
canceled		Manually canceled by admin or user.

Viewing All Subscriptions

Navigate to Admin Panel > Subscriptions tab. The table displays every warehouse with:

- Warehouse name and location
- Current plan name and tier
- Subscription status (color-coded badge)
- Expiry date (current_period_end)

Warehouses without a subscription show "No Plan" with no status badge.

Manually Editing a Subscription

1. Click the pencil icon (or Edit Subscription on mobile) for the target warehouse.
2. The edit dialog opens with two tabs: Details and Payments.
3. On the Details tab, set:
 - Plan -- select from the dropdown of all available plans
 - Status -- choose: Active, Incomplete, Past Due, Canceled, Unpaid, or Trialing
 - Expiry Date -- the date when the subscription period ends
4. Click Save Changes.

Activation Safeguards

The system applies automatic safeguards when you save:

- Setting status to active with a future expiry date: Clears any grace period data and resets the grace_period_notified flag. This handles manual renewals during a grace period.
- Setting status to grace_period: Automatically calculates a grace_period_end date 7 days after the expiry date.
- Changing from grace_period to any non-expired status: Clears all grace period data.

Warning: Setting a subscription to active without an expiry date means it has no automatic expiration. The warehouse will remain on the selected plan indefinitely until manually changed.

Subscription Code Generation

Activation codes allow offline or manual subscription distribution. Codes follow the format XXXX-XXXX-XXXX using an unambiguous character set (no I/1/O/0 confusion).

Generating Codes

1. Go to Admin Panel > Codes tab.
2. Click Generate Codes.
3. In the dialog, configure:
 - Plan -- which plan the code activates
 - Duration (Days) -- how many days the subscription lasts after redemption
 - Quantity -- number of codes to generate (1-50)
 - Notes (Internal) -- optional reference like "Bulk order for Client X, Payment Ref #123"
4. Click Generate.
5. The generated codes appear in a list. Click Copy All to copy them to clipboard.

Code Lifecycle

Status | Meaning

available | Code is valid and can be redeemed

used | Code has been redeemed by a warehouse

revoked | Code was manually disabled by admin

Code Redemption (User Side)

When a warehouse owner redeems a code:

1. The code is validated against the subscription_codes table.
2. If valid and available, it is marked as used with optimistic locking (WHERE status = 'available').
3. The subscription is upserted for the warehouse with the code's plan and duration.
4. If the warehouse already has an active subscription with a future expiry date, the new duration is added to the existing end date (stacking).

Warning: If the subscription upsert fails after the code is marked as used, the code is burned. The error message instructs the user to contact support with the code string. Monitor error logs for redeemCodeAction_subUpdate failures.

Handling Expired Subscriptions

Subscription expiry is processed automatically by a cron job / edge function that calls processExpiredSubscriptions(). The lifecycle is:

active (past expiry) --> grace_period (7 days) --> expired --> downgrade to Free

Grace Period

When a subscription's `current_period_end` passes:

1. Status changes to `grace_period`.
2. A `grace_period_end` is set to 7 days after the original expiry.
3. An in-app notification is sent to the warehouse: "Your subscription expired. You have X days to renew."
4. The `grace_period_notified` flag prevents duplicate notifications.

Expiry and Downgrade

When the grace period ends:

1. Status changes to `expired`.
2. The subscription is downgraded to the Free plan.
3. A downgrade notification is sent: "Your subscription has been downgraded to the Free plan."

Proactive Expiry Warnings

A daily cron job calls `sendExpiryWarnings()` to notify warehouses before expiry. Warnings are sent at:

- 7 days before expiry
- 3 days before expiry
- 1 day before expiry

Manual Expiry Processing

In emergencies or for testing, super admins can trigger expiry processing manually. This is restricted to the `super_admin` role and calls the same `auto_expire_subscriptions` database function.

Razorpay Payment Integration

Subscription payments are processed through Razorpay payment links:

1. A payment link is created with the plan price and warehouse owner's details.
2. The link is sent via SMS to the owner's phone number.
3. The link expires after 3 days.
4. On successful payment, a webhook activates the subscription automatically.

The webhook handler performs idempotency checks (prevents double-activation) and validates the payment amount against the plan price with a tolerance of 1 rupee.

After activation, a confirmation SMS is sent to the owner with the plan name and validity period.

Chapter 22: Roles and Permissions

GrainFlow uses Role-Based Access Control (RBAC) to govern what each user can see and do. Roles are assigned at two levels: a global profile role (on the profiles table) and a warehouse-level role (on the warehouse_assignments table).

Role Definitions

Role | Scope | Description

super_admin | Global | Platform operator. Full access to all warehouses and the Admin Panel.

owner | Warehouse | Warehouse owner. Full control over their warehouse(s), billing, and team. Can also access the Admin Panel.

admin | Warehouse | Warehouse administrator. Can manage staff, customers, storage, and settings. Cannot manage billing.

manager | Warehouse | Operational manager. Can handle day-to-day storage, customers, and payments. Limited settings access.

staff | Warehouse | Basic operator. Can view and create records but cannot delete or manage team.

customer | External | External party (farmer/business). No application login -- represented as data records only.

Permission Matrix

The following table shows access for each role across all application modules. Permissions are: Full (read + write + delete), Write (read + write), Read (view only), or None (no access).

Dashboard and Navigation

Module | super_admin | owner | admin | manager | staff

Main Dashboard | Full | Full | Full | Read | Read

Admin Panel | Full | Full | None | None | None

Warehouse Switcher | Full | Full | None | None | None

Customer Management

Action | super_admin | owner | admin | manager | staff

View customers | Full | Full | Full | Full | Read

Create customer | Full | Full | Full | Full | None

Edit customer | Full | Full | Full | Full | None

Delete customer | Full | Full | Full | None | None

Storage Records

Action		super_admin		owner		admin		manager		staff
View records		Full		Full		Full		Full		Read
Create inflow		Full		Full		Full		Full		Write
Process outflow		Full		Full		Full		Full		None
Bulk outflow		Full		Full		Full		None		None
Edit record		Full		Full		Full		Write		None
Delete record		Full		Full		Full		None		None
Export records		Full		Full		Full		Full		None

Payments

Action		super_admin		owner		admin		manager		staff
View payments		Full		Full		Full		Full		Read
Record payment		Full		Full		Full		Full		None
Send payment link		Full		Full		Full		Full		None
Send SMS reminder		Full		Full		Full		Full		None
Delete payment		Full		Full		None		None		None

Expenses

Action		super_admin		owner		admin		manager		staff
View expenses		Full		Full		Full		Full		None
Create expense		Full		Full		Full		Write		None
Edit expense		Full		Full		Full		None		None
Delete expense		Full		Full		None		None		None

Reports and Analytics

Action		super_admin		owner		admin		manager		staff
View reports		Full		Full		Full		Read		None
Export reports (PDF/Excel)		Full		Full		Full		Read		None
Analytics dashboard		Full		Full		Full		None		None
Platform analytics		Full		Full		None		None		None

Settings

Action		super_admin		owner		admin		manager		staff
View warehouse settings		Full		Full		Full		None		None
Edit warehouse settings		Full		Full		Write		None		None
Manage team (add/remove users)		Full		Full		Full		None		None
Change user roles		Full		Full		Full		None		None

Notification preferences	Full	Full	Full	Full	Full
SMS settings	Full	Full	Full	None	None
Billing / Subscription	Full	Full	None	None	None

Admin Panel (super_admin and owner only)

Action	super_admin	owner	admin	manager	staff
View all warehouses	Full	Full	None	None	None
Create warehouse	Full	Full	None	None	None
View all users	Full	Full	None	None	None
View audit logs	Full	Full	None	None	None
View analytics	Full	Full	None	None	None
Manage subscriptions	Full	Full	None	None	None
Generate activation codes	Full	None	None	None	None
Process expired subscriptions	Full	None	None	None	None

How Roles Are Assigned

Global Role (profiles.role)

Set when a user account is created. The super_admin role is assigned directly in the database and should not be changed through the UI. The owner role is typically set during warehouse onboarding.

Warehouse Role (warehouse_assignments.role)

Assigned when a user is granted access to a specific warehouse. A user can have different roles in different warehouses.

To change a warehouse role:

1. Go to Settings > Team in the target warehouse.
2. Find the user in the team list.
3. Select the new role from the dropdown.
4. The change takes effect immediately.

All role changes are recorded in the audit log with the UPDATE action on the USER entity.

Role Enforcement Layers

Roles are enforced at three layers:

1. UI Layer: Navigation items and action buttons are conditionally rendered based on the user's role. Users

never see UI elements they cannot use.

2. Server Action Layer: Every server action ('use server' function) verifies the authenticated user's role before performing any operation. Unauthorized requests return an error.

3. Database Layer (RLS): Row Level Security policies on Supabase ensure that even if the UI and server action layers are bypassed, the database will not return data the user is not authorized to see. See Chapter 25 for details.

Subscription-Based Limits

In addition to role-based permissions, subscription tier limits are checked before certain actions:

- Adding a team member: The system checks the USERS limit for the warehouse's plan before allowing a new warehouse_assignments record.
- Creating a storage record: The system checks the STORAGE_RECORDS limit.
- Accessing gated features: Features like Analytics Dashboard or API Access require a minimum plan tier.

When a limit is reached, the user sees an "Upgrade Required" message with the feature name and the minimum tier needed.

Chapter 23: Notifications and SMS

GrainFlow provides two notification channels: in-app notifications (the notification center) and SMS messages via the TextBee integration. Both channels can be configured per-warehouse.

In-App Notifications

Notification Center

The notification center is accessible from the bell icon in the application header. It displays notifications scoped to the current warehouse and the current user.

Notifications have the following properties:

Field	Description
Title	Short headline (e.g., "Payment Received")
Message	Detailed notification body
Type	info, warning, error, or success -- controls the visual style
Category	Grouping label: subscription, payment, storage, system
Read status	Tracked per-user via the notification_reads table

Managing Notifications

- Mark as read: Click on a notification to mark it as read. The read state is tracked in a separate notification_reads join table keyed on (notification_id, user_id).
- Mark all as read: Use the "Mark all as read" action to bulk-update all unread notifications. This processes in chunks of 1,000 for performance.

Notification Preferences

Each user can configure which notification types they receive per warehouse. Navigate to Settings > Notifications to toggle:

Setting	Description	Default
Payment Received	Notify when a payment is recorded	On
Low Stock Alert	Notify when stock levels are low	On
Pending Dues	Notify about overdue payments	On
New Inflow	Notify when new storage is created	On
New Outflow	Notify when withdrawal occurs	On

Each setting can be toggled independently for three delivery channels:

Channel	Description
In-App	Shows in the notification center
Email	Sends to the user's registered email
SMS	Sends via TextBee (requires SMS to be enabled)

System-Generated Notifications

The following notifications are generated automatically:

Event	Type	Category
Subscription expiring in 7/3/1 days	warning	subscription
Subscription entered grace period	warning	subscription
Subscription downgraded to Free	error	subscription
Subscription activated via payment	success	subscription

SMS Integration (TextBee)

GrainFlow uses TextBee as its SMS gateway. TextBee works by routing SMS through an Android device registered to your TextBee account, making it cost-effective for Indian phone numbers.

Prerequisites

Before SMS will work, you need:

1. A TextBee account (sign up at textbee.dev)
2. The TextBee Android app installed on a device
3. The device registered and connected in your TextBee dashboard

Environment Configuration

Set the following environment variables in your `.env.local` (development) or in your hosting provider's environment settings (production):

Variable	Description	Example
TEXTBEE_API_KEY	Your TextBee API key from the dashboard	tb_api_xxxxxxx
TEXTBEE_DEVICE_ID	The device ID from your TextBee dashboard	64a1b2c3d4e5f6

Both values are required. If either is missing, the service logs a warning on startup and all SMS operations return an error.

SMS Permission Model

SMS is not available to all warehouses by default. Access is controlled through a layered permission check:

1. Environment whitelist (highest priority): Set SMS_ALLOWED_USERS or SMS_ALLOWED_WAREHOUSES as comma-separated IDs in environment variables to grant access regardless of plan.
2. Plan feature flag: If the warehouse's subscription plan has allow_sms: true in its features JSON, SMS is enabled.
3. Default: If neither condition is met, SMS is blocked with the message "SMS service is disabled for trial users."

Per-Warehouse SMS Settings

Each warehouse can toggle specific SMS message types on or off. Navigate to Settings > SMS to configure:

Setting	Description	Default
Payment Reminders	Send SMS when payment is overdue	On
Inflow Welcome	Send SMS when new storage is created	Off
Outflow Confirmation	Send SMS when withdrawal is processed	Off
Payment Confirmation	Send SMS when payment is received	Off

These settings are stored in the sms_settings table per warehouse. Default settings are created automatically on first access.

SMS Templates

All SMS messages follow a structured format. The TextBee service formats messages automatically. Phone numbers are validated to 10 digits (Indian format) with the +91 prefix added automatically.

Payment Reminder

{WarehouseName}
Payment Due: Rs.{Amount}
Record: {RecordNumber}
Customer: {CustomerName}
Please clear dues at earliest.

Payment Confirmation

{WarehouseName}
Payment Received: Rs.{Amount}
Record: {RecordNumber}
Thank you, {CustomerName}!

Inflow Welcome (Storage Started)

{WarehouseName}

Storage Started

Record: {RecordNumber}

Customer: {CustomerName}

Item: {Commodity}

Bags: {BagCount}

Location: {Location}

Date: {StorageDate}

Outflow Confirmation (Withdrawal)

{WarehouseName}

Withdrawal Confirmed

Invoice: {InvoiceNumber}

Record: {RecordNumber}

Customer: {CustomerName}

Item: {Commodity}

Bags Withdrawn: {BagCount}

Rent: Rs.{RentAmount}

Hamali: Rs.{HamaliAmount}

Total: Rs.{TotalAmount}

Thank you!

Drying Finalization

{WarehouseName}

Drying Finalized

Record: {RecordNumber}

Customer: {CustomerName}

Item: {Commodity}

Final Bags: {BagCount}

Hamali: Rs.{HamaliAmount}

Stock Updated.

Subscription Payment Link

Upgrade to {PlanName} (Rs.{Price}). Pay: {PaymentLink}

- {BusinessName}

Subscription Activation Confirmation

{PlanName} activated! Valid until {ExpiryDate}

- {BusinessName}

Long Message Handling

If an SMS exceeds 160 characters, the TextBee service automatically splits it into multiple parts. Each part is suffixed with a counter (e.g., (1/2), (2/2)) and sent sequentially with a 500ms delay between parts.

Bulk SMS

The service supports bulk sending through `sendBulkSMS()`, which processes all recipients in parallel using `Promise.allSettled()`. Individual failures do not block other messages.

Troubleshooting SMS

See Chapter 24 for common SMS issues and their solutions.

Chapter 24: Troubleshooting

This chapter covers common issues administrators encounter, organized by symptom, cause, and fix.

Login and Authentication

User cannot log in

Symptoms: User sees "Invalid credentials" or is redirected back to the login page repeatedly.

Cause: The user's Supabase Auth account may not exist, the password may be wrong, or the session cookie may be corrupted.

Fix:

1. Verify the user exists in the Supabase Auth dashboard (Authentication > Users).
2. If the account exists, send a password reset email from the Supabase dashboard.
3. If the session seems stuck, have the user clear browser cookies for the application domain and try again.
4. Check that the NEXT_PUBLIC_SUPABASE_URL and NEXT_PUBLIC_SUPABASE_ANON_KEY environment variables are correctly set in production.

User logs in but sees an empty dashboard

Symptoms: The user authenticates successfully but the dashboard shows no data -- no customers, no records, no warehouse name.

Cause: The user's profiles record may not have a warehouse_id set, or they may not have an active entry in warehouse_assignments.

Fix:

1. Check the profiles table for the user's row -- verify warehouse_id is populated.
2. Check the warehouse_assignments table for an active assignment (where deleted_at IS NULL).
3. If the assignment was soft-deleted, restore it via Settings > Team in the target warehouse or by using toggleWarehouseAccess().

Missing Records (Row Level Security)

User reports data is missing but it exists in the database

Symptoms: A user says they cannot see certain customers, storage records, or payments. A super_admin can see the same data without issues.

Cause: Row Level Security (RLS) policies restrict data access based on the user's warehouse assignment. If a user's warehouse_id does not match the record's warehouse_id, the database returns zero rows -- not an error.

Fix:

1. Confirm the user's warehouse assignment in the warehouse_assignments table.
2. Confirm the records in question belong to the same warehouse_id.
3. If the user was recently moved between warehouses, they will only see data for their current warehouse.
4. RLS never throws an error for missing access -- it silently returns no rows. This is by design to prevent data leakage between tenants.

Warning: Do not disable RLS policies to "fix" missing data issues. Instead, correct the user's warehouse assignment. Disabling RLS would expose all tenant data system-wide.

Newly created record does not appear

Symptoms: A user creates a customer or storage record but it does not show in their list immediately.

Cause: The record may have been created with the wrong warehouse_id, or the page cache has not revalidated.

Fix:

1. Hard-refresh the page (Ctrl+Shift+R / Cmd+Shift+R).
2. If still missing, check the database directly -- look for the record and verify its warehouse_id matches the user's assignment.
3. Check that the revalidatePath() call is present in the server action that created the record.

Permission Errors

User gets "Forbidden" or "You do not have permission"

Symptoms: User clicks an action button and sees a permission error toast or a 403 page.

Cause: The user's role does not have the required permission for that action. This check happens both in the UI (button visibility) and in the server action (authorization check).

Fix:

1. Check the user's role in `warehouse_assignments` for the current warehouse.
2. Refer to the permission matrix in Chapter 22 to confirm whether the role should have access.
3. If the role should have access, check for a bug in the server action's authorization logic.
4. To change the user's role: go to Settings > Team, find the user, and select the appropriate role from the dropdown.

Super admin sees 403 on Admin Panel

Symptoms: A user who should be a `super_admin` gets the "403 Forbidden" page when navigating to `/admin`.

Cause: The Admin Panel checks `profiles.role` (the global profile role), not `warehouse_assignments.role`. The user may have admin or owner as their warehouse role but not `super_admin` or owner on their profile.

Fix:

1. Check the profiles table directly: `SELECT role FROM profiles WHERE id = '<user_id>'`.
2. If the role is not `super_admin` or owner, update it directly in the database: `UPDATE profiles SET role = 'super_admin' WHERE id = '<user_id>'`.
3. The user must log out and log back in for the profile role change to take effect.

Payment Link Failures

Payment link creation fails

Symptoms: Clicking "Send Payment Link" shows an error toast. No SMS is sent.

Cause: Multiple possible causes in the payment link creation chain.

Fix: Check each step in order:

1. Razorpay credentials: Verify `RAZORPAY_KEY_ID` and `RAZORPAY_KEY_SECRET` are set in environment variables.
2. Owner phone number: The system fetches the warehouse owner's phone from `profiles.phone` via the `user_warehouses` join. If the owner has no phone number, the link creation fails with "Owner phone number not available."
3. Plan configuration: Verify the target plan exists in the plans table and has a valid price value.
4. Razorpay API errors: Check the error logs (Sentry) for the operation `createSubscriptionPaymentLink`. Razorpay may be rejecting the request due to invalid amount, expired API keys, or account issues.

Payment completed but subscription not activated

Symptoms: The customer paid via the Razorpay link, but the subscription still shows as "incomplete" or "unpaid."

Cause: The webhook from Razorpay may not have been received or processed successfully.

Fix:

1. Check Sentry for errors tagged with operation activateSubscriptionPayment.
2. Verify the webhook URL is correctly configured in the Razorpay dashboard (should point to /api/razorpay/webhook).
3. Verify the RAZORPAY_WEBHOOK_SECRET environment variable matches the secret configured in Razorpay.
4. Check for idempotency: if the subscription_payments table already has a record for this razorpay_payment_id, the payment was already processed.
5. As a workaround, manually activate the subscription through Admin Panel > Subscriptions -- edit the warehouse, set the plan, status to active, and set the expiry date.

SMS Not Sending

SMS fails with "SMS service is disabled for trial users"

Symptoms: Attempting to send any SMS returns this error message.

Cause: The warehouse does not have SMS permission. SMS requires either an environment whitelist entry or a plan with allow_sms: true.

Fix:

1. Add the warehouse ID to the SMS_ALLOWED_WAREHOUSES environment variable (comma-separated).
2. Or add the user's ID to SMS_ALLOWED_USERS.
3. Or ensure the warehouse's subscription plan includes "allow_sms": true in its features JSON column.

SMS fails with "TextBee Device ID not configured"

Symptoms: SMS operations fail with this specific error.

Cause: The TEXTBEE_DEVICE_ID environment variable is not set.

Fix:

1. Log in to your TextBee dashboard at textbee.dev.
2. Navigate to Devices and find your registered Android device.
3. Copy the Device ID.
4. Add it to your environment: `TEXTBEE_DEVICE_ID=<your_device_id>`.
5. Restart the application for the change to take effect.

SMS fails with "Invalid phone number"

Symptoms: SMS returns "Invalid phone number. Must be 10 digits."

Cause: The customer's phone number in the database is not a valid 10-digit Indian number. The system strips the +91 prefix and non-digit characters before validation.

Fix:

1. Edit the customer record and correct the phone number to a valid 10-digit number.
2. Avoid storing country codes or special characters in the phone field -- the system adds +91 automatically.

SMS sent but customer did not receive it

Symptoms: The system reports SMS sent successfully (no error), but the recipient did not get the message.

Cause: The TextBee Android device may be offline, out of SMS credits, or have poor network connectivity.

Fix:

1. Check the TextBee dashboard for the message delivery status.
2. Verify the Android device is powered on, connected to the internet, and has mobile network signal.
3. Check the `sms_logs` table for the message record and its status.
4. Try sending a test SMS from the TextBee dashboard directly to rule out API issues.

Subscription Not Activating

Code redemption fails with "Invalid code"

Symptoms: A warehouse owner enters an activation code and gets "Invalid code."

Cause: The code may be mistyped, already used, or not in the database.

Fix:

1. Check the `subscription_codes` table for the code string (codes are case-insensitive -- the system converts to uppercase).
2. Verify the code's status is available. If it is used or revoked, it cannot be redeemed again.
3. Have the admin generate a new code if needed.

Code redeemed but subscription shows wrong end date

Symptoms: After redeeming a code, the subscription end date is earlier than expected.

Cause: If the warehouse had no active subscription (or an expired one), the duration starts from today. If it had an active subscription with a future end date, the duration is added to the existing end date. The issue may be that the admin expected stacking but the current subscription was already expired.

Fix:

1. Check the `subscriptions` table for the warehouse's `current_period_end`.
2. If incorrect, manually adjust via Admin Panel > Subscriptions -- edit the warehouse and set the correct expiry date.

Performance and General Issues

Admin Panel loads slowly

Symptoms: The Admin Panel takes several seconds to render.

Cause: The Admin Panel loads all warehouses, all users, analytics data, all subscriptions, and all codes in a single server-side render. On large platforms, this can be slow.

Fix:

1. This is expected for platforms with many warehouses. The data is fetched server-side so the user sees a complete page on load.
2. Monitor Sentry performance traces for the `/admin` route to identify the slowest query.
3. Consider adding database indexes if specific queries are slow.

Audit log shows "unknown" IP address

Symptoms: Some audit log entries show "unknown" instead of an IP address.

Cause: The x-forwarded-for and x-real-ip headers were not present in the request. This can happen with certain proxy configurations or when actions are triggered by cron jobs / background processes.

Fix:

1. For web requests, ensure your hosting provider (Vercel, etc.) forwards the x-forwarded-for header.
2. For cron-triggered actions, "unknown" is expected since there is no client IP.

Chapter 25: Security

This chapter explains GrainFlow's security architecture in practical terms for administrators responsible for the system.

Multi-Tenant Data Isolation

GrainFlow is a multi-tenant system where multiple warehouses share the same database. Data isolation is the most critical security property: Warehouse A must never see Warehouse B's customers, records, or payments.

This isolation is enforced at the database level using Row Level Security (RLS).

How Row Level Security Works

Every major table in the database (storage_records, customers, payments, expenses, etc.) has a warehouse_id column. Supabase RLS policies are attached to these tables that automatically filter queries based on the logged-in user's warehouse assignment.

In simple terms:

1. A user logs in and gets an auth session with their user ID.
2. When they query any table, the database checks: "Which warehouses is this user assigned to?"
3. The database only returns rows where warehouse_id matches one of the user's assigned warehouses.
4. This filtering is invisible to the application code -- it happens inside PostgreSQL before results are returned.

This means even if a bug in the application code forgets to add a WHERE warehouse_id = ... clause, the database will still enforce the filter. This is a defense-in-depth strategy.

What RLS Does NOT Do

- RLS does not filter based on role (owner vs. staff). Role-based restrictions within a warehouse are handled at the application layer.
- RLS does not apply to database admin connections or service role keys. The Supabase service role key bypasses all RLS policies. Never expose this key to the client.

Warning: The SUPABASE_SERVICE_ROLE_KEY bypasses all RLS. It is used only in server-side code for admin operations (cron jobs, webhooks). It must never be exposed to the browser or included in NEXT_PUBLIC_ environment variables.

Authentication

GrainFlow uses Supabase Auth with server-side rendering (SSR). Key security properties:

- Session management: Auth sessions are stored in HttpOnly cookies, not localStorage. This prevents JavaScript-based session theft (XSS attacks).
- CSRF protection: Built into the Supabase SSR auth flow.
- Server-side verification: Every server action re-verifies the user's auth session before processing. There is no trust of client-side session claims.

Session Flow

1. User submits credentials to Supabase Auth.
2. Supabase returns a session token stored in an HttpOnly cookie.
3. On each server action call, the server reads the cookie, verifies the token with Supabase, and retrieves the current user.
4. If the token is invalid or expired, the user is redirected to /login.

RBAC Enforcement

Role-Based Access Control is enforced at three layers, as described in Chapter 22. From a security perspective:

1. UI layer hides controls but is not a security boundary (can be bypassed with browser dev tools).
2. Server action layer is the primary enforcement point. Every 'use server' function checks the user's role before performing operations.
3. Database layer (RLS) is the last line of defense, ensuring data access is constrained even if the server action layer has a bug.

The authenticatedAction Pattern

Many server actions use a wrapper function called `authenticatedAction` that:

1. Retrieves the current user from the auth session.
2. Fetches the user's profile and role.
3. Passes the verified user, Supabase client, and role to the action handler.
4. Returns an error if authentication fails.

This centralizes auth checking and prevents individual actions from forgetting to verify the session.

Audit Trail

All significant operations are recorded in the `audit_logs` table. This provides a tamper-evident record of who did what, when, and from where.

Audit Log Structure

Field | Description

id | Unique log entry identifier

warehouse_id | Which warehouse the action affected

user_id | Who performed the action

action | CREATE, UPDATE, DELETE, LOGIN, LOGOUT, EXPORT, BULK_ACTION

entity | STORAGE_RECORD, CUSTOMER, PAYMENT, INFLOW, OUTFLOW, USER, SETTINGS, SUBSCRIPTION

entity_id | The specific record affected

details | JSON with action-specific context (e.g., old values, new role, operation type)

ip_address | The actor's IP address (from x-forwarded-for or x-real-ip headers)

created_at | Timestamp of the action

Immutability

The audit log schema defines no Update type -- entries are write-once. The application never updates or deletes audit log rows. This ensures the trail cannot be tampered with through normal application operations.

Non-Blocking Design

Audit logging is designed to never block the primary operation. If the audit log insert fails (network issue, database error), the failure is logged to Sentry but the main action still succeeds. This prevents audit logging bugs from breaking business operations.

Soft Deletes

GrainFlow uses soft deletes throughout the system instead of hard deletes. When a record is "deleted":

1. The deleted_at column is set to the current timestamp.
2. The record remains in the database but is filtered out of normal queries.
3. Application queries include WHERE deleted_at IS NULL (or equivalent RLS policy) to exclude soft-deleted records.

This applies to:

- Warehouses (warehouses.deleted_at)
- Warehouse assignments (warehouse_assignments.deleted_at)
- Storage records (storage_records.deleted_at)

Benefits

- Audit integrity: The audit trail references entity IDs that still exist in the database.
- Recovery: Accidentally deleted records can be restored by setting deleted_at back to NULL.

- Compliance: Data retention requirements can be met without losing records.

Warning: Soft-deleted records still count toward storage usage in the database. If a warehouse has many soft-deleted records, they consume space but are not visible to users. Periodic cleanup of very old soft-deleted records should be done directly in the database with caution.

Input Validation

All user inputs are validated using Zod schemas before reaching the database:

- String sanitization: The `sanitizeString` utility removes `<` and `>` characters to prevent XSS injection.
- Phone number validation: Phone numbers are stripped of non-digit characters and validated to 10 digits.
- Email validation: Standard email format validation via Zod.
- Numeric validation: Amounts, quantities, and IDs are type-checked and range-validated.

React's built-in JSX escaping provides an additional layer of XSS prevention. The application does not use `dangerouslySetInnerHTML`.

Rate Limiting and API Security

Server Actions

GrainFlow uses Next.js Server Actions instead of traditional API routes for most operations. Server Actions are:

- Automatically authenticated (cookie-based)
- Type-safe with TypeScript
- Validated with Zod schemas
- Not directly callable as REST endpoints

Webhook Security

The Razorpay webhook endpoint (`/api/razorpay/webhook`) verifies the webhook signature using the `RAZORPAY_WEBHOOK_SECRET` before processing any payment event. Unverified webhooks are rejected.

Payment Security

- Payment amounts are validated against plan prices with a tolerance of 1 rupee to prevent amount manipulation.
- Idempotency checks prevent double-processing of the same payment (checked via `razorpay_payment_id`).

Encryption

In Transit

All connections use TLS (HTTPS). Supabase enforces TLS 1.3 for database connections. The application is deployed on Vercel which enforces HTTPS for all traffic.

At Rest

Data at rest is encrypted by Supabase's infrastructure. The database storage, backups, and connection strings are all encrypted. Sensitive environment variables are managed through the hosting provider's encrypted environment variable storage.

Secrets Management

Secret | Storage | Never Expose To

SUPABASE_SERVICE_ROLE_KEY | Server env only | Browser, client code, logs

RAZORPAY_KEY_SECRET | Server env only | Browser, client code

RAZORPAY_WEBHOOK_SECRET | Server env only | Browser, client code

TEXTBEE_API_KEY | Server env only | Browser, client code

SENTRY_AUTH_TOKEN | Server env only | Browser, client code

NEXT_PUBLIC_SUPABASE_URL | Client-safe | N/A (public by design)

NEXT_PUBLIC_SUPABASE_ANON_KEY | Client-safe | N/A (public, RLS protects data)

Warning: Never commit .env.local to version control. The .gitignore file should include this entry. Rotate secrets immediately if they are accidentally exposed.

Error Handling and Logging

Centralized Error Logger

All errors are logged through the `logError()` utility, which:

1. Sends the error to Sentry with structured context (operation name, user ID, warehouse ID, metadata).
2. Logs to the console in development mode.
3. Never exposes stack traces or internal details to the end user.

Error Messages

User-facing error messages are generic and do not reveal implementation details. For example:

- "An unexpected error occurred" instead of database error messages
- "Unauthorized" instead of describing which permission check failed
- "Failed to send SMS" instead of exposing TextBee API error responses

This prevents information leakage that could help an attacker understand the system's internals.

Security Monitoring

Sentry Integration

All runtime errors are captured by Sentry with:

- Tags: Operation name for filtering (e.g., activateSubscriptionPayment)
- User context: User ID (when available)
- Extra data: Warehouse ID and action-specific metadata

Recommended Monitoring Practices

1. Review Sentry alerts daily for unusual error patterns.
2. Run npm audit monthly to check for dependency vulnerabilities.
3. Review the audit log periodically for unusual admin actions.
4. Monitor the Supabase dashboard for unusual query patterns or auth failures.
5. Rotate API keys and secrets quarterly.

Security Checklist for Administrators

- ☐ All environment variables are set correctly in production
- ☐ SUPABASE_SERVICE_ROLE_KEY is not exposed in any NEXT_PUBLIC_ variable
- ☐ HTTPS is enforced on the production domain
- ☐ Supabase RLS is enabled on all tables (verify in Supabase dashboard > Authentication > Policies)
- ☐ Sentry error monitoring is active and alerting
- ☐ Database backups are enabled in Supabase
- ☐ TextBee device is online and connected
- ☐ Razorpay webhook secret matches the dashboard configuration
- ☐ .env.local is in .gitignore and not committed
- ☐ Monthly npm audit is scheduled

Glossary

Terms used throughout the GrainFlow application and this manual.

Storage & Inventory

Term | Definition

Inflow | Stock coming into the warehouse. Each inflow creates a storage record.

Outflow | Stock leaving the warehouse. Can be single-record or bulk.

Bags | The standard unit of measure for commodities. One bag is one physical sack of grain/commodity.

Storage Record | A single entry tracking one batch of commodity for one customer, from inflow to outflow.

Lot | A physical zone or area within the warehouse (e.g., Row A, Section B, Shed 3).

Commodity | The type of grain or product stored (e.g., Paddy, Wheat, Red Gram, Cotton).

FIFO | First-In, First-Out. The allocation strategy used for bulk outflow, where the oldest stock is withdrawn first.

Bulk Outflow | Withdrawing bags across multiple storage records in a single operation.

Financial

Term | Definition

Hamali | Labor charges for loading and unloading bags at the warehouse. Charged per bag.

Rent | Storage fee charged based on duration and number of bags. Tiered pricing: Rs. 36/bag for 0-6 months, Rs. 55/bag for 7-12 months.

Khata | An initial advance or credit amount recorded at the time of inflow.

Total Due | The sum of all unpaid rent and hamali charges for a customer.

Payment Allocation | Distributing a single payment across multiple storage records, typically using FIFO.

Discount | A reduction applied to the calculated rent during outflow.

Users & Roles

Term | Definition

Super Admin | System-wide administrator with access to all warehouses and the admin panel.

Owner | The warehouse owner with full access to their warehouse operations and settings.

Admin | A warehouse administrator with full operational access but no system-level admin.

Manager | A warehouse manager with read access to reports and analytics but limited write access.

Staff | A warehouse worker with basic inflow, outflow, and payment recording capabilities.

Customer | An end customer who stores goods. Has read-only portal access to view their own records.

Subscription & Billing

Term | Definition

Plan Tier | The subscription level: Free, Starter, Professional, or Enterprise.

Grace Period | A 7-day window after subscription expiry before downgrade to Free tier.

Activation Code | A code generated by super admins that customers can redeem to activate a plan.

Payment Link | A Razorpay-generated URL sent to customers for online payment.

Technical

Term | Definition

RLS | Row Level Security. A database feature that ensures users can only see data from their assigned warehouses.

Warehouse Isolation | The principle that data from one warehouse is never visible to users of another warehouse.

Soft Delete | Marking a record as deleted without physically removing it from the database. Allows recovery.

Audit Trail | An automatic log of all changes made in the system, including who changed what and when.

Webhook | An automated notification from Razorpay to GrainFlow when a payment status changes.